

KJ Growth data engineer



Schole.ai

Founding growth engineer
at an AI startup



trykindredsong.com

Sold custom pet songs
Revenue run rate: 20k



Sponsorfi.io

Built a SaaS tool
A/B tested signup flow



Dropshipper

LED lights + hair trimmers
Revenue run rate: 40k



USC Marshall

Undergrad

Code changes

D1–D16: verified defects and how each was proven

#	Defect	How it was proven
D1	Google Ads signup conversion has never fired: it listens for new_user_signed_up, the app sends signup	No emitter in ~1,090 files; sealed replay shows signup sends nothing to Google
D2	Each platform gets a different "purchase": LinkedIn no value/ID, X twice with random IDs, Meta and TikTok first-purchase-only valued at LTV, business_subscription consumed by nothing	Sealed replay of the app's real purchase pushes, decoded per vendor
D3	Renewals, upgrades, add-ons, refunds reach no platform from the browser	Shipped code: these events only refresh the screen
D4	Fallback value bug: if the invoice lookup fails, annual plans report at half the monthly price with a Date.now() ID	Shipped code, PRICES table and the fallback builder
D5	TikTok signup event has an empty event_id	Sealed replay
D6	Google form_start/form_submit fire on every Generate click; Meta and X fire auto click events	Logged-in capture of a real generation
D7	Meta loses the click on multi-page and return journeys; no code builds _fbclid from the stored fbclid	Live click-through matrix: one-hop keeps it, multi-hop and return lose it
D8	ChatGPT-ads oppref always lost on marketing landings	Live test; the SDK reads only the current URL
D9	iOS Google IDs (gbraid/wbraid) and Reddit's rdt_cid never reach the server store; checkout posts only gclid	Shipped code: the app keeps 4 keys and overwrites the JSON
D10	In-app browser handoff copies a bare /home URL; Google OAuth stays enabled where Google blocks it	Live Instagram/TikTok UA tests, overlay inspected
D11	uBlock's default list strips all 7 click IDs and blocks every pixel plus their own /4vu8/ path	Replayed with uBlock's actual filter lists
D12	Click IDs live in script-set cookies Safari caps at 7 days	Cookie inspection; only oa_device_id is server-set
D13	Page views: Google/Reddit/LinkedIn/X get 0 on route changes, Meta up to 3, UET 3, TikTok 2, legacy Amplitude 6	Logged-in route-change counts
D14	No consent manager, no Consent Mode (gcd=13l3l3l3l1l1 on all 348 Google hits), 0 of 18 tags gated	Live hits, DE/GB/CH probes
D15	Enterprise HubSpot form hard-codes lead_source=Event, lead_source_detail=Brandweek; click-ID fields never filled	Public form definition JSON
D16	Shim logs click IDs to console, ttclid cookie lasts 1 day, quantity=undefined in checkout URL, GTM loaded twice, Clear-Site-Data on the homepage	Code and captures

Ned the Salmon

The seal	Every request to a collection endpoint (Google Ads, their /4vu8/ paths, facebook.com/tr, the CAPI Gateway, TikTok, Reddit, LinkedIn, X, Bing, Amplitude, Clarity, Hotjar, Tolt, OpenAI) is intercepted, recorded and failed locally. The browser can reach only openart.ai; third-party scripts are fetched by the watchdog itself; prefetch, prerender, service workers, WebSockets, popups and the Reporting API are disabled; profiles are deleted after each run. Every report includes the zero-leak proof (1,181 and 1,172 attempts blocked, 0 completed).
19 contract checks	Each is mapped to a defect: signup reaches Google with user data; TikTok signup has reg_<uid>; X sends one deterministic purchase; LinkedIn has value and ID; business_subscription is consumed; no auto form events on generation; one page view per route change on 7 platforms; _fbclid present after multi-hop and return; oppref preserved; gbraid/wbraid persisted; the handoff URL carries attribution; consent defaults set for EEA/UK/CH.
Replay journeys	Journeys replayed with synthetic click IDs: Meta one-hop, multi-hop, return; oppref landing; gbraid/wbraid; Instagram in-app browser; a uBlock-stripped landing; in-app route changes; consent probes for DE, GB, CH.
Patch inputs	For the "after" run: --patch-container and --patch-gtag-config (serve modified GTM/Google tag files for both loaders), --inject-script (a stand-in for the drop-in shim), --edge-sim (applies the Worker's cookies to responses).
Output	results.json and a red/green report.html with evidence, per-platform click-ID coverage, consent summary, container diff and the leak proof.
Scheduling	Dockerfile, Cloud Run Job + Scheduler, Slack alert on any failure or container change, GitHub Actions alternative. Not deployed.

Edge attribution

withAttribution(handler) wraps the fetch handler of their existing Cloudflare Worker.

Capture + cookies	On document navigations, parse 14 click-ID types plus UTMs, referrer and landing path. Set a signed, HttpOnly 13-month oa_attr with first and last touch, server-set oa_ad_clids in their existing format, _fbc minted per Meta's rules, __oppref, and a 28-day ttclid (D7, D8, D9, D11, D12). Server-set cookies address Safari's script-set expiry limit. Browser privacy rules still apply.
Persistence	Persist to KV keyed by device ID without blocking the response, or POST to an extension of their existing /api/user/ad-click-ids. Signup-merge logic ships as runnable code.
In-app handoff	POST /api/attribution/handoff creates a single-use 30-minute token holding only allow-listed params and a same-origin path. Credential-looking paths are refused. GET /r/:token restores cookies in the external browser (D10).
Consent	Outside EEA/UK/CH, eligible unless GPC, a US opt-out or an explicit denial applies. Inside EEA/UK/CH, advertising storage needs an explicit grant.
Blocked users	On a uBlock-stripped landing, save UTMs and surviving IDs. Together with hashed-email server conversions, these support attribution of blocked users at campaign level.
Parity + security	dist/edge-sim.js is byte-consistent with the Worker for Ned the Salmon. Security review fixed a ReDoS in bot detection and a forged-device-ID path. All 45 hand-broken rules were caught.

Contracts

The shared foundation for events, IDs, consent and scoring

Canonical events	12 canonical events: signup, activation_first_generation, checkout_started, purchase_first, purchase_renewal, purchase_upgrade, purchase_add_on, purchase_one_time_pack, refund, chargeback, enterprise_lead and lead_stage_change. JSON Schema, TypeScript types and zod validators include a compile-time drift guard.
IDs + hashing	Reuse the browser's exact IDs: reg_<uid>, purchase_<invoiceld>, sub_<invoiceld>, and Reddit's SHA-256 of the ID. Per-platform hashing verified on 3,305 cases.
Platform mapping	An 84-row event mapping per platform with requires_web_fix flags.
Consent + scoring	The consent region list and GPC/opt-out fields. The PurchaseValueScore contract.
Source schemas	Source schemas shaped like their real Stripe, credit-ledger and Amplitude data.
Fixtures + costs	Synthetic fixtures, 109 model costs at list price, and a deterministic 2,000-user cohort generator with labelled illustrative parameters.

Conversion service

A Cloud Run service built with node:http

Inputs	Stripe webhooks with offline signature verification: checkout, invoice paid for create/cycle/update, subscription updated/deleted, refunds and disputes. POST /events handles signup, activation, checkout started, enterprise lead and lead stage. Pub/Sub is an alternative.
Pipeline	Normalise events, write the BigQuery ledger, resolve value and click IDs (building fbc from stored fbclid), then re-check consent at send time. An idempotent outbox per (platform, event ID) has backoff, dead-letter, per-document isolation, lease renewal and parking of out-of-order Stripe events.
Senders	Google Data Manager API uses the tag's transactionId. Meta CAPI uses the same event IDs, external_id lower-cased like the browser, system_generated for renewals and a 7-day guard. Senders also cover TikTok, Reddit, LinkedIn, X and Microsoft.
Refunds	Google restated with a phantom-conversion guard. Microsoft restated or retracted. The rest skipped and logged.
Value + Meta	Use the purchase-time profit score, else cash tagged cash_fallback. Record floors. A value-health command checks Meta's thresholds. GET /value lets their invoice lookup return profitValueMajor so the pixel sends the same number.
Retention + mode	Retention TTLs and POST /tasks/erase. Dry-run by default. Live mode refuses in-memory stores and example secrets.
Offline test	The end-to-end test replays 27 Stripe and 10 app events with zero network calls.

The fish tank

43 dbt models compiling for both BigQuery and DuckDB

Core marts	fct_conversion_ledger: one row per event, browser-format IDs, refunds linked via invoice-payment and the purchase's own keys. fct_data_quality_quarantine. fct_generation_cost: ledger × cost seed, discount variable. fct_user_features_24h: time-bounded.
Profit scores	fct_predicted_profit_24h: signup+24h unconditional, for experiments only. fct_purchase_value_score: purchase-time conditional, the ad value. Both have append-only logs and fitted-parameter snapshots for reproducibility.
Experiments + audiences	fct_experiment_profit_by_arm and fct_experiment_profit_by_arm_daily: intention-to-treat, per day × segment × slice, model-error variance, p·V–C decomposition and CUPED covariate. fct_audience_candidates.
Reconciliation	fct_reconciliation splits every Ads-vs-Stripe gap by cause: window, date basis, time zone, refunds, double counting, LTV inflation, blocked users and fallback.
Data handling	Currency exponent, FX and tax handling. Subscription status from the invoice line's period. PII whitelisted at staging. Aggregate-only Metabase views.
Synthetic backtest	Out-of-time backtest on synthetic data: purchase-time score error \$10.93 versus \$19.96 for first-charge revenue, calibration slope 1.08 and 79% interval coverage. Pipeline proof.
Sizing + review	Week-1 sizing queries against their real schema. Five Metabase questions. 27 of 29 injected bugs caught.

Bandit allocator

Replaces hand-tuned default-model flags edited 28 and 27 times

Sampling	Thompson sampling per arm per segment, rewarded on the 24h profit signal decomposed as $p \cdot V - C$ for lower variance.
Traffic guards	5% exploration floor, ± 10 percentage points/day, a never-touched holdout, and activation/refund/failure guardrails.
Statistical guards	A sequentially valid stop-loss. In simulation, false alarms on null arms fell from 65–78% to 0–2%. A fail-closed traffic-split check against logged weights. An MDE-based minimum sample.
Adaptation	A 14-day half-life, inverse-propensity weighting, resets on model or promo changes, and warm starts for new arms.
Output	A LaunchDarkly semantic patch per flag through their approval workflow. No app code change.
Simulation	Illustrative simulation with 10 replications: any bandit beats the hand-tuned split. Profit vs conversion is not distinguishable on synthetic data.

My first 90 days

What I can confirm from outside and what needs your systems

Week 1	Verify with your own data and change nothing. Run the sizing queries, confirm the dead signup conversion in Google Ads, and schedule the watcher.
Weeks 2–6	Ship the fix pack, the edge module, a consent platform and the form fix. Land the ledger and the reconciliation.
Weeks 7–12	Turn on server-side conversions at cash value. Calibrate the profit value on real data, then prove it with a lift test.
Confirmed from outside	Every tracking finding above.
Needs your systems	How Stripe webhooks are handled today. What already lands in BigQuery. Real per-model costs. EU consent behavior. Ad-platform API access.

Reviews and totals

Four independent reviews found roughly 60 issues; fixes have regression coverage

1,484

reported TypeScript tests

505

dbt checks

Major fixes verified

Firestore reliability + URL validation

Profit scored at the correct moment

Stop-loss with fewer false alarms

27 of 29 injected database bugs caught

Sealed tracking proof rerun on final code

8 packages built. Deployment and real-data validation remain.